

Artificial Intelligence – AT2

SIMATS ENGINEERING
COURSE NAME: Artificial Intelligence
COURSE CODE: CSA17

Q1. Which of the following best describes a Logical Agent in AI?

- a) An agent that reacts randomly to the environment
 - b) An agent that uses a knowledge base and logical inference to decide actions
 - c) An agent that only follows pre-defined scripts
 - d) An agent that learns from rewards
- b) An agent that uses a knowledge base and logical inference to decide actions.**

Small Explanation: A logical agent uses a knowledge base and logical inference to choose suitable actions.

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- a) $P \wedge \neg Q$
 - b) $\neg P \vee Q$
 - c) $\neg Q \rightarrow \neg P$ only
 - d) Both b and c
- d) Both b and c.**

Small Explanation: $P \rightarrow Q$ is equivalent to $\neg P \vee Q$ and also to its contrapositive $\neg Q \rightarrow \neg P$.

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- a) A disjunction of conjunctions
 - b) A conjunction of disjunctions (clauses)
 - c) A conjunction of literals only
 - d) A sequence of implications
- b) A conjunction of disjunctions (clauses).**

Small Explanation: CNF is an AND of clauses, where each clause is an OR of literals.

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- a) Some students pass
 - b) Every entity that is a student passes
 - c) If something passes, it is a student
 - d) There exists at least one student who passes
- b) Every entity that is a student passes.**

Small Explanation: The universal quantifier means the rule applies to every entity that is a student.

Q5. Unification in FOL is the process of finding substitutions that:

- a) Convert clauses to CNF
- b) Make two logical expressions syntactically identical

- c) Prove a goal using backward chaining
- d) Derive facts using Modus Ponens

b) Make two logical expressions syntactically identical.

Small Explanation: Unification finds substitutions that make two logical expressions match.

Q6. In the Resolution algorithm, deriving the empty clause ■ indicates:

- a) The knowledge base is consistent
- b) The negated goal is satisfiable
- c) A contradiction — the original goal is proven
- d) More clauses need to be added

c) A contradiction — the original goal is proven.

Small Explanation: The empty clause represents a contradiction, proving the goal by resolution refutation.

Q7. Forward Chaining is best described as a _____ strategy.

- a) Goal-driven / top-down
- b) Data-driven / bottom-up
- c) Heuristic-based depth-first
- d) Probabilistic inference

b) Data-driven / bottom-up.

Small Explanation: Forward chaining starts from known facts and derives new facts using rules.

Q8. Backward Chaining begins reasoning from:

- a) All known facts toward a conclusion
- b) The goal and works backward to verify supporting facts
- c) A random clause in the knowledge base
- d) The resolution refutation tree

b) The goal and works backward to verify supporting facts.

Small Explanation: Backward chaining is goal-driven and searches backward for supporting facts.

Q9. Which step is NOT part of converting a sentence to CNF?

- a) Eliminate biconditionals ($\leftrightarrow$)
- b) Move negations inward using De Morgan's laws
- c) Apply Skolemization to remove existential quantifiers
- d) Distribute AND over OR

c) Apply Skolemization to remove existential quantifiers.

Small Explanation: Note: In full FOL-to-CNF conversion, Skolemization is normally used when existential quantifiers are present; this question is wording-sensitive.

Q10. Knowledge Engineering in FOL involves:

- a) Training a neural network on domain-specific data

- b) Defining domain concepts, relations, and axioms formally in FOL
- c) Writing heuristic search strategies
- d) Encoding reward functions for reinforcement learning

b) Defining domain concepts, relations, and axioms formally in FOL.

Small Explanation: Knowledge engineering formally represents domain knowledge using concepts, relations, facts, and rules.

Q11. Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:

- a) $\neg Q$ is true
- b) Q must be true
- c) P is uncertain
- d) The KB is inconsistent

b) Q must be true.

Small Explanation: From $P \rightarrow Q$ and P, Modus Ponens allows us to conclude Q.

Q12. The Most General Unifier (MGU) is the substitution that:

- a) Is the most specific possible match
- b) Makes two terms identical with minimal variable binding
- c) Eliminates all variables from a clause
- d) Applies resolution to produce CNF

b) Makes two terms identical with minimal variable binding.

Small Explanation: The MGU makes two terms identical using the least restrictive variable bindings.

Q13. Which inference method is more appropriate when a specific goal is known?

- a) Forward Chaining
- b) Truth Table method
- c) Backward Chaining
- d) A* search

c) Backward Chaining.

Small Explanation: Backward chaining starts with the specific goal and searches for facts that support it.

Q14. Propositional Logic differs from First Order Logic because it:

- a) Supports probabilistic reasoning
- b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions
- c) Is always more expressive
- d) Handles uncertainty better

b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions.

Small Explanation: Propositional logic represents propositions as true or false, while FOL represents objects and relationships.

Q15. Resolution Refutation proves a goal G by:

- a) Directly deriving G from the KB using forward chaining
- b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause ■)
- c) Checking G in the truth table
- d) Applying Modus Tollens repeatedly

b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause ■).

Small Explanation: If the negation of G produces a contradiction, the original goal G is proven.

Q16. Learning from observation in AI refers to:

- a) An agent receiving numerical rewards from the environment
- b) Improving performance by examining examples or experience from the environment
- c) Acquiring knowledge by following explicit symbolic rules
- d) Memorizing a fixed dataset without generalization

b) Improving performance by examining examples or experience from the environment.

Small Explanation: The system learns useful patterns or knowledge from observations and experience.

Q17. Inductive learning is best described as:

- a) Deriving specific conclusions from general rules
- b) Generalizing a hypothesis from specific training examples
- c) Memorizing training data exactly
- d) Applying search algorithms to find optimal policies

b) Generalizing a hypothesis from specific training examples.

Small Explanation: Inductive learning moves from specific examples to a general rule or hypothesis.

Q18. In a Decision Tree, what does each internal node represent?

- a) A class label / output decision
- b) A probability distribution over outcomes
- c) A test on an attribute / feature
- d) A training data sample

c) A test on an attribute / feature.

Small Explanation: Each internal node tests a feature and uses the result to select a branch.

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

- a) Requires large amounts of training data to generalize
- b) Learns only from numeric data using regression
- c) Uses prior domain knowledge to explain and generalize a single example
- d) Applies unsupervised clustering to find hidden patterns

c) Uses prior domain knowledge to explain and generalize a single example.

Small Explanation: EBL uses existing domain knowledge to explain an example and form a general rule.

Q20. In statistical learning, a Naive Bayes classifier assumes:

- a) All features are conditionally dependent on each other given the class
- b) The data must follow a uniform distribution
- c) All features are conditionally independent given the class label
- d) Labels are determined by a reward function

c) All features are conditionally independent given the class label.

Small Explanation: Naive Bayes makes the simplifying assumption that features are conditionally independent given the class.

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

- a) Using the EM algorithm to estimate missing values
- b) Training a deep neural network with dropout regularization
- c) Directly computing maximum likelihood estimates from observed data
- d) Applying reinforcement signals to update weights

c) Directly computing maximum likelihood estimates from observed data.

Small Explanation: When all variables are observed, model parameters can be estimated directly from the data.

Q22. The Expectation-Maximization (EM) algorithm is used when:

- a) Training data is fully labeled and no variables are hidden
- b) An agent explores an unknown environment using trial and error
- c) The dataset contains hidden (latent) variables that cannot be directly observed
- d) Backpropagation fails to converge in a neural network

c) The dataset contains hidden (latent) variables that cannot be directly observed.

Small Explanation: EM estimates model parameters when some variables are hidden or the data is incomplete.

Q23. In a Neural Network, the activation function serves to:

- a) Initialize the weights of each neuron to zero
- b) Normalize the input dataset before training
- c) Introduce non-linearity to enable learning of complex patterns
- d) Store previously learned examples in memory

c) Introduce non-linearity to enable learning of complex patterns.

Small Explanation: Non-linear activation functions allow neural networks to learn complex relationships.

Q24. Which of the following correctly describes backpropagation in a neural network?

- a) Weights are updated in the forward pass using input gradients
- b) Outputs of each layer are stored and reused in later epochs
- c) The error is propagated backward from output to input layers to adjust weights
- d) A separate model is trained for each hidden layer independently

c) The error is propagated backward from output to input layers to adjust weights.

Small Explanation: Backpropagation calculates error gradients and uses them to update network weights.

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

- a) Minimizing the number of training examples used
 - b) Grouping similar data points into clusters using centroids
 - c) Finding the maximum-margin hyperplane that separates classes
 - d) Updating weights using Q-values from an environment
- c) Finding the maximum-margin hyperplane that separates classes.**

Small Explanation: SVM finds a separating hyperplane with the largest possible margin between classes.

Q26. The kernel trick in kernel machines (SVMs) allows:

- a) Reducing the number of support vectors needed
 - b) Accelerating gradient descent convergence
 - c) Operating in a high-dimensional feature space without explicitly computing coordinates
 - d) Replacing hidden layers in a deep neural network
- c) Operating in a high-dimensional feature space without explicitly computing coordinates.**

Small Explanation: The kernel trick computes similarities as if data were mapped to a higher-dimensional space.

Q27. In Reinforcement Learning, the agent aims to maximize:

- a) Classification accuracy on a fixed test set
 - b) The number of features selected from the training data
 - c) Cumulative reward received over time through interaction with the environment
 - d) The size of the neural network hidden layer
- c) Cumulative reward received over time through interaction with the environment.**

Small Explanation: Reinforcement learning aims to maximize long-term cumulative reward.

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

- a) A supervised learning algorithm requiring labeled datasets
 - b) An unsupervised clustering technique using centroids
 - c) A model-free, off-policy temporal difference learning method
 - d) A decision-tree-based ensemble classifier
- c) A model-free, off-policy temporal difference learning method.**

Small Explanation: Q-learning learns action values without requiring a model of the environment.

Q29. Overfitting in machine learning occurs when:

- a) The model performs poorly on both training and test data
- b) The learning rate is set too low during gradient descent
- c) The model fits training data too closely and generalizes poorly to new data
- d) The training set has more features than training examples

c) The model fits training data too closely and generalizes poorly to new data.

Small Explanation: An overfitted model performs well on training data but poorly on unseen data.

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

- a) RL uses labeled input-output pairs; SL uses reward signals
- b) Both RL and SL require explicit reward functions
- c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples
- d) RL is always faster to converge than SL on any dataset

c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples.

Small Explanation: Reinforcement learning learns from rewards during interaction, while supervised learning uses labeled examples.